

JAVA 8 FEATURES ASSIGNMENT 2

Q1. Employee Bonus Eligibility Check

Concepts: Predicate, Stream, Optional

Write a program that filters a list of employees based on their salary using a Predicate. If eligible, retrieve and print their email addresses using Optional. If the email is missing, print "Email not available".

Input : List.of(new Employee(name,salary, email), ...so on);

Q2. Lazy Configuration Loader

Concepts: Supplier, Optional

Assume you have a configuration setting that may or may not be loaded. Use a Supplier to lazily load the config value **only if it's missing**. Use Optional to simulate missing or present values and print the final result.

Hint : Create a Supplier loadConfig which will have a configuration string.

Q3. Product Discount Application

Concepts: Function, Optional, Method Reference, Map

Create a discount system using a Function that applies a 10% discount to a product price. If the price is available (use Optional), apply the discount and print the final price. Otherwise, print 0.0.

Hint: Function<T,R> calculate the price in function.

Q4. Customer Notification System

Concepts: Consumer, Stream API

Create a list of customers with a boolean field indicating whether they are subscribed. Use Stream API and a Consumer to send notifications **only to subscribed customers**.

Input : List.of(new Customer(name,subscribed),so on);

Q5. User Search and Email Transformation

Concepts: Optional, Method Reference, Function, Stream

From a list of users, find a user by name (use **Stream**) and get their email address (use **Optional**). Convert the email to **uppercase** using a **Function** and print it if present.

Input: List.of(new User(name, email),...more user so on.);